

Model Portability for EMT Studies of IBR

This white paper introduces the need for network model management in electromagnetic transient (EMT) studies. A standard data format would help users build, validate, update, and simulate EMT models to keep up with increasing demand for EMT studies of inverter-based resources (IBR). The existing “base case” methods work for steady-state and dynamic network models in planning and operations, but these methods require customized, ad-hoc extensions for EMT modeling [1]. Two approaches may be considered to solve this problem. The first approach extends the IEC standard Common Information Model (CIM), which has been successful in Europe with steady-state and dynamics modeling, but slow to adopt in North America. The second approach begins with a text format that is widely used to maintain “base cases” in North America, extending that for EMT applications in an IEEE open-source software project. The open-source approach may incorporate some ideas that were used in CIM, but with more emphasis on ease and speed of adoption.

Table of Contents

EMT Model Repository Requirements.....	1
Balance-of-Network Model Components.....	3
Plant Model Components.....	6
CIM for Dynamics and EMT.....	8
CIM Extensions for Transformer Saturation.....	8
CIM Extensions for Exact IBR Control Code.....	9
Future Work for CIM Implementation.....	12
Future Work for IEEE Open-Source Implementation.....	12
References.....	15
Scope from EMT TF WI3 Meeting Minutes.....	17
Revision History.....	17

EMT Model Repository Requirements

As IBRs appear in greater numbers on the bulk electric system (BES), NERC has encouraged utilities to perform EMT studies of IBR [2]. This creates a new burden on the engineers to create and maintain robust EMT models of many IBR plants, along with extensive BES network models for EMT. Network models for power flow (PF) and positive sequence phasor domain transient (PSPDT)¹ applications are maintained in tool-specific file formats that have become de facto standards. However, PF and PSPDT models don’t have enough detail for EMT studies, so their de facto standard file formats don’t suffice for EMT. As with all power system studies, the EMT model provides the foundation for all processes and results, and building the model can be the most time-

¹ A more precise term for “stability” analysis, under consideration by the NERC EMT Task Force and the IEEE P2800.2 Working Group.

consuming phase of an EMT study. This white paper summarizes approaches to reduce the effort and increase the quality of EMT model-building.

Currently, engineers must assemble, verify, and fill gaps in EMT model components manually, which is both time intensive and error prone. This process is semi-automated, based on PF/PSPDT files [3], with specific gaps in zero-sequence data, geographic information, transformer and bus mismatches, boundary identification and equivalencing, validating solutions, dynamic load modeling, and synchronizing updates [1], [4]. Instead of storing links to an EMT model of IBR plants, one could store links to a standardized model. This approach has shown success with CIM-based EMT model building in France [5] and in Europe for HVDC [6]. Proprietary solutions have been used in Japan [7].

Figure 1 shows how the *CIM Network and Dynamics* schema [8]², [9] can support a framework for building and maintaining EMT models for IBR studies. The *Planning Tool Export* comes from PF, PSPDT, and/or short-circuit (SCKT) native file formats. The *Asset and Settings Data* may define control and protection settings, line construction, transformer test reports, etc., for more EMT details. At present, *User Edits* are necessary to complete the EMT model, but with appropriate extensions, CIM could lessen the need for such manual interventions.

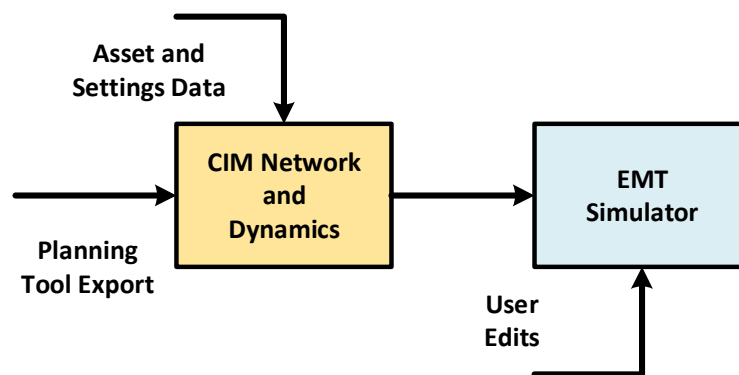


Figure 1: Collecting power system data into a model for electromagnetic transients [10]

Figure 2 shows more detail of the framework to implement a repository of EMT network and IBR plant models. The network model is usually based on bus-branch PSPDT and SCKT data. The *Other* data may include protective device settings, node-breaker topologies, and extensions to be determined later. EMT simulators also require *Initial Conditions* to begin time stepping, so the network model should support different sets of steady-state solutions to provide these initial conditions. In the utility workflow, this network model may be updated at three-to-six-month intervals. There may also be different versions of the network model for different planning scenarios. Under different guises, the CIM already supports PSPDT, SCKT, bus-branch, node-breaker, initial conditions, and planning scenarios.

However, the CIM does not currently support a repository of IBR models. The utility maintains its own network model, but various project developers and hardware vendors provide most of the IBR data. The IBR models include custom parameters, custom logic, and increasingly, custom binary control code in the form of a dynamic link library (DLL). Some of the PSPDT tools support these in

² This link was taken down late in 2024, possibly due to intellectual property concerns of IEC and/or ENTSO-E.

the form of “user code”, but not in a standardized way, and not in a way that supports EMT. As with the network model, IBR models must be maintained in different versions and validated.

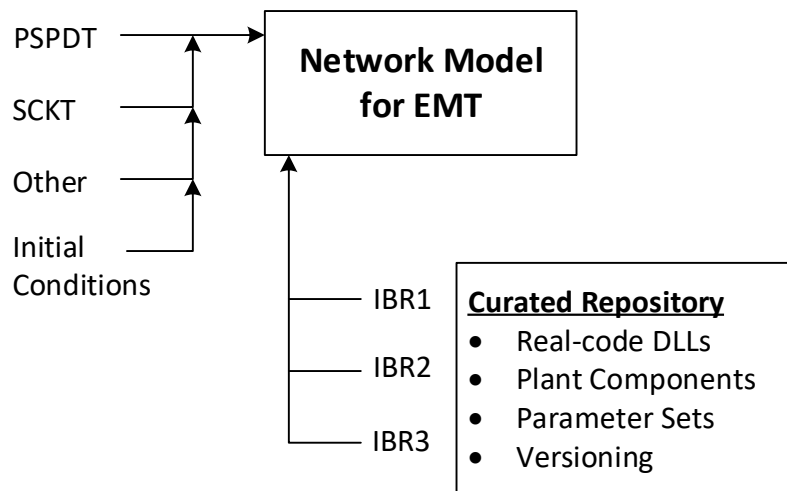


Figure 2: Attributes of an EMT Network and IBR Model Repository [11]

Simulation results from a preliminary CIM-based framework have been obtained using the Alternative Transients Program (ATP) with two medium-scale BES networks [12]:

- **IEEE118** – representative of the Midwestern United States circa 1962, with 56 machines aggregating steam and hydro generators, 14 solar generators, and 5 wind generators. Includes 193 buses considering generator step-up (GSU) transformers [13].
- **WECC240** – representative of the Western Electricity Coordinating Council (WECC) system, with 49 machines aggregating steam and hydro generators, 25 solar generators, and 10 wind generators. Includes two series capacitors, two aggregations of distributed energy resources (DER), and 243 buses [14].

Data and script files for these examples are available to licensed ATP users [15].

The next several sections of this white paper present some heuristics to get started with a CIM-based EMT modeling framework, based on PF and/or PSPDT data, i.e., no zero-sequence or asset-based information. Minimum extensions for transformer saturation and IBR plant models are then presented. The white paper concludes with two options for implementation, one adopting the CIM, and the other creating an IEEE open-source project.

Balance-of-Network Model Components

EMT supports frequency-dependent and unbalanced line models, which require data on the conductors, bundling, tower cross section, and midspan sags for overhead lines, or geometric and materials information for underground lines. However, frequency-dependent models are not needed for IBR studies, as the frequency range of interest is rather low compared to other EMT applications. Unbalanced line models are not needed for IBR studies because the dynamics are primarily positive sequence. It’s necessary to represent single-line-to-ground faults (SLGF) accurately, but this can be done without using unbalanced line models. For EMT studies of IBR, it’s

sufficient to use balanced (or transposed) line models with 60-Hz resistance, inductance, and capacitance values for positive and zero sequence.

The CIM *ACLineSegment* class has positive sequence parameters r , x , and bch in SI units, zero-sequence parameters $r0$, $x0$, and $bch0$ in SI units, and a parent class *Conductor.length* attribute in meters. Per-unit impedances from PF or PSPDT models must be converted to SI units for CIM. Both line charging parameters, bch and $bch0$, must be specified as non-zero. The zero-sequence data may be available from SCKT network models. If not, it can be estimated for transmission lines:

$$L = x / (\text{length} * \omega) \quad (1)$$

Where ω is 377 for 60-Hz systems. If $bch > 0$, estimate the capacitance and surge impedance:

$$C = bch / (\text{length} * \omega) \quad (2)$$

$$Z = \sqrt{L/C} \quad (3)$$

Otherwise, assuming a typical overhead line surge impedance, estimate bch with:

$$Z = 400 \quad (4)$$

$$C = L/Z^2 \quad (5)$$

$$bch = C * \text{length} * \omega \quad (6)$$

Verify the positive sequence (line mode) propagation velocity does not exceed the speed of light:

$$v = 1/\sqrt{LC} \leq 3e8 \quad (7)$$

If v is 50% or less of $3e8$, then the line is probably underground cable, not overhead.

If $Z < 100$ in (3), assume the line is underground cable and estimate zero-sequence parameters by:

$$r0 = r \quad (8)$$

$$x0 = x \quad (9)$$

$$b0ch = bch \quad (10)$$

Otherwise, assume overhead line and estimate zero-sequence parameters as follows, using either the PF/PSPDT value for bch , or the value estimated from (6):

$$r0 = 2r \quad (11)$$

$$x0 = 2x \quad (12)$$

$$b0ch = 0.6 * bch \quad (13)$$

If the line is known to have bundled sub-conductors, then $Z=300$ may be used in (4). If the branch consists of two or more lines in parallel, the surge impedance, Z , should be reduced accordingly. If the original PF/PSPDT model included line charging data for the branch, then (7) should still help distinguish between overhead and underground lines. If the original data did not include line charging data, then it's probably an overhead line. Cable line charging values are higher and more important than for overhead lines, hence more likely to have been maintained in PF/PSPDT models.

Most of the EMT line models for IBR studies should include traveling wave behaviors. The tool may refer to these as Bergeron, Clarke, Karrenbauer or distributed transposed models. However, if the line is so short that its travel time is less than the simulation time step, a single pi section may be used to represent it. The CIM-to-EMT exporter could make this choice based on a user-supplied Δt .

Figure 3 shows important considerations for transformer modeling. Legacy PF models may still represent three-winding transformers with separate two-winding transformers in a star equivalent (Figure 3 left) because legacy PF tools only represented two-winding transformers. One of the legacy two-winding transformers is likely to have negative reactance, which is okay for PF simulations but not for EMT. Furthermore, the star point is retained in such legacy PF models with a base voltage of, typically, 1 kV or 0.1 kV. In preparing such models for EMT, the two-winding transformers must be reaggregated into correct three-winding models.

To represent the transformer zero-sequence behavior for EMT, it's essential to correctly specify the delta or wye connection type for each winding. It's also important to correctly specify any neutral impedance, or ungrounded connection, for wye windings. This data is typically not included with PF or PSPDT models.

Transformer saturation can also be important to model in EMT studies of IBR, because under dynamics conditions, saturation may increase the reactive power³ demands on generators. For IBR studies, this can be modeled with sufficient accuracy by a two-slope characteristic (Figure 3 right). It's possible to use more points and slopes, but the data is not readily available.

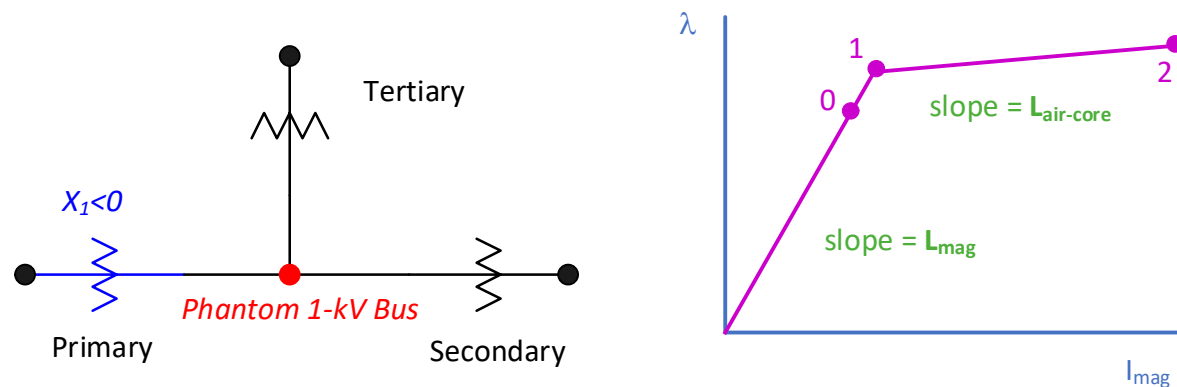


Figure 3: Three-winding transformer star equivalent (left) and saturation characteristic (right) for EMT [12]

EMT saturation models are generally defined in characteristics of instantaneous flux linkage, λ , vs. instantaneous magnetizing current, I_{mag} . The CIM *TransformerCoreAdmittance* class has a *b* attribute for magnetizing susceptance, and a *g* attribute for core losses, both in SI units. If these are not available from PS/PSPDT data, then typical non-zero values should be assumed, e.g., 1% magnetizing current and 0.25% core loss current, then convert those to SI values. Use the secondary, or lowest-voltage winding, excluding the tertiary, for connecting the saturation branch. The short-circuit inductance, *x*, between primary and secondary windings is also needed, with *x* in ohms referred to the secondary winding. The secondary winding's rated voltage, *u*, should be expressed in peak volts, not RMS volts. The steady-state magnetizing current, I_o , should be

³ It's not rigorous to view transformer saturation current as reactive power, but the practice is widespread.

expressed in peak amps, not RMS amps, referred to the secondary winding. From this information, we can estimate the slopes and point **0** in Figure 3 (right) as follows:

$$L_{mag} = b/\omega \quad (14)$$

$$L_{air-core} = 2x/\omega \quad (15)$$

$$\lambda_0 = u/\omega \quad (16)$$

Then we choose a typical “knee voltage”, V_{knee_pu} , for the transformers, typically 1.05 to 1.15 per-unit. This defines a straight-line extrapolation from point 0 to point 1 in Figure 3 (right). Then determine points **1** and **2** as follows:

$$I_1 = V_{knee_pu} * I_0 \quad (17)$$

$$\lambda_1 = V_{knee_pu} * \lambda_0 \quad (18)$$

$$\Delta I = 1000 \quad (19)$$

$$I_2 = I_1 + \Delta I \quad (20)$$

$$\lambda_2 = \lambda_1 + L_{air-core}\Delta I \quad (21)$$

EMT saturation models typically take points **1** and **2** for input, with the origin implied. The steady-state solution may use point **0**. In the linear range of operation, this will be consistent with points 1 and 2 after time-stepping begins.

Plant Model Components

Figure 4 shows the rotating machine connected to a single-machine infinite bus (SMIB) test system at extra-high voltage (EHV). The plant is connected at a point of common coupling (PCC). The generator model in dq axis form, a single inertia, J ,⁴ and typical control systems are highlighted in orange. PSPDT models, and the CIM for Dynamics schema, include these items. The plant also minimally includes a GSU transformer, highlighted in green. The GSU data is generally available in PSPDT models, but the delta winding connection should be specified for EMT. The CIM already supports the items in Figure 4.

Figure 5 shows an IBR plant model connected to the SMIB at a PCC. The balance-of-plant model, highlighted in green, includes more components than in Figure 4, but they are all supported in CIM. The IBR-specific components, highlighted in orange, are not presently supported in CIM. Extensions to enable this support will be discussed next.

The *Exact or Generic Controls* in Figure 5 may comprise a real-code DLL as defined in [16], with an example in [17]. It may also comprise a WECC generic model, which is already supported in CIM. Other options include custom user-models, which would be specific to the EMT tool and not addressed in this white paper. Figure 6 summarizes the DLL interface, which may be incorporated

⁴ One of the early EMT applications was to perform sub-synchronous resonance (SSR) studies of generators in series-compensated BES. For SSR studies, the single inertia, J , had to be replaced with a multi-mass rotational system. That data was not available in PSPDT models, which is still usually the case.

into the CIM by reference, especially if the DLL interface is formalized as an IEEE standard⁵. The orange-highlighted converter bridge and passive filter components in Figure 5 might be specified in a tractable number of attributes for a new CIM class, as discussed later. EMT tools generally have their own inverter bridge modeling options, including average sources for either exact or generic controls, or pulse-width-modulated (PWM) switching topologies for exact controls.

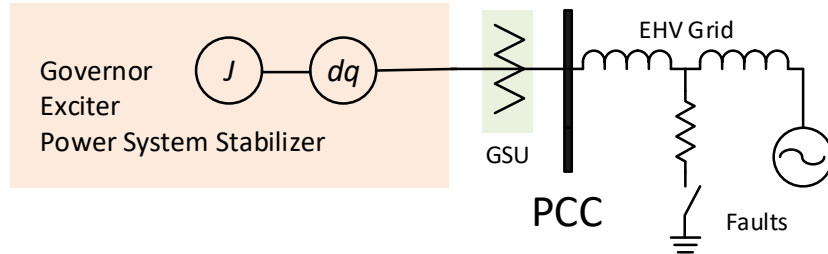


Figure 4: Rotating machine plant model components connected to a SMIB test system [18]

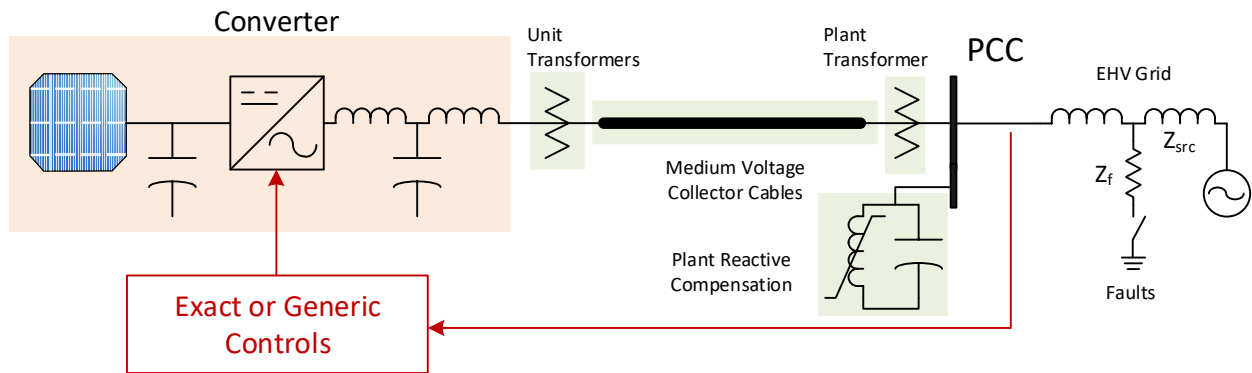


Figure 5: IBR plant model components connected to a SMIB test system

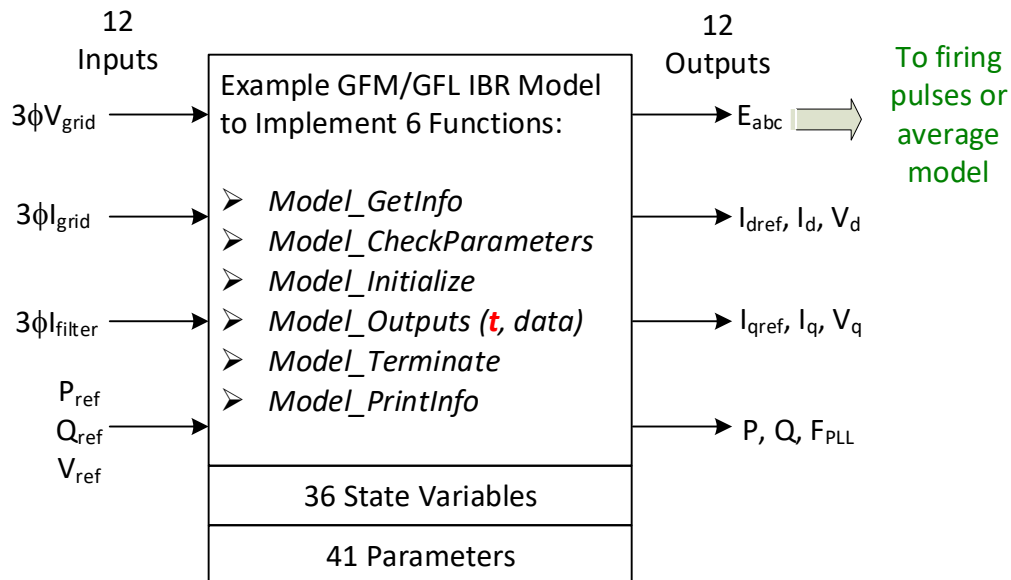


Figure 6: DLL interface for example IBR control code models of grid-forming (GFM) and grid-following (GFL) inverters [18]

⁵ The model user must understand the meaning of 12 inputs and 12 outputs in Figure 6. These may vary in other examples that meet the DLL specification.

CIM for Dynamics and EMT

Figure 7 shows the current CIM Dynamics support for WECC models, including some of the newer DER aggregation models, in the form of a unified modeling language (UML) schema. In CIM, the inverter is called a *PowerElectronicsConnection*, which is part of the *Wires* package. It has a *Terminal* for connection to other grid components, e.g., the *Unit Transformers* from Figure 5. A similar pattern is used for connecting exciter, governor, and power system stabilizer dynamics to a *SynchronousMachine* instead of *PowerElectronicsConnection*. No further extensions are needed to support EMT studies with WECC models, as demonstrated in [12].

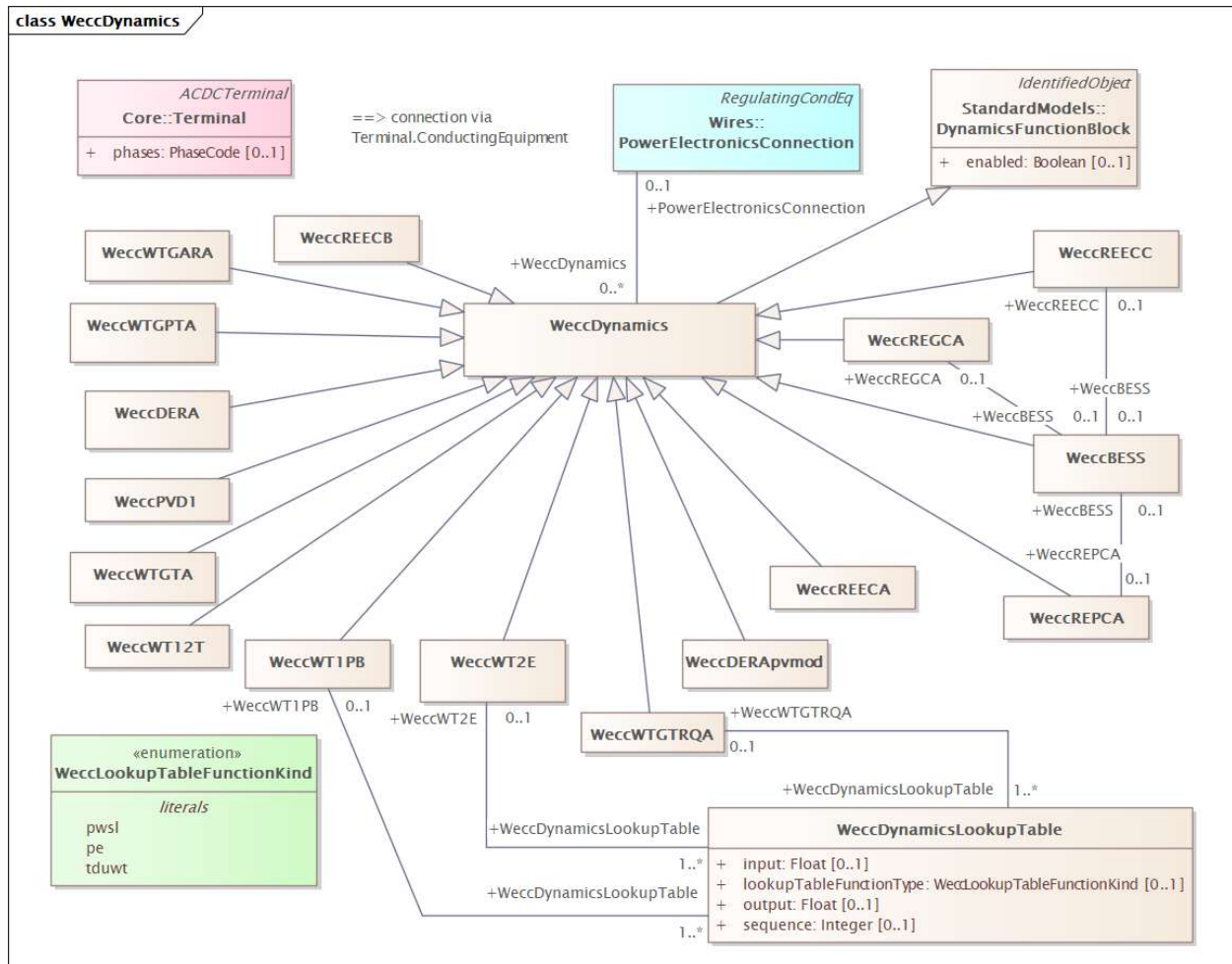


Figure 7: CIM Dynamics UML for the WECC Generic IBR Models

CIM Extensions for Transformer Saturation

Figure 8 shows UML for a *TransformerSaturationCurve* with attributes defined in Table 1. It uses a well-established pattern in CIM, by associating an optional *TransformerSaturationCurve* to the *TransformerCoreAdmittance*. There will be two instances of *CurveData* for each saturation characteristic, defining points **1** and **2** of Figure 3 (right) with equations (17), (18), (20), and (21).

Table 1: CIM Attributes for Transformer Saturation Curve

Name	CIM Type	Description
Curve		
xUnit	UnitSymbol	A (current)
y1Unit	UnitSymbol	Vs (flux linkage, choosing Vs over Wb)
CurveData		
xvalue	Float	Instantaneous current, I _{mag}
y1value	Float	Instantaneous flux linkage, lambda

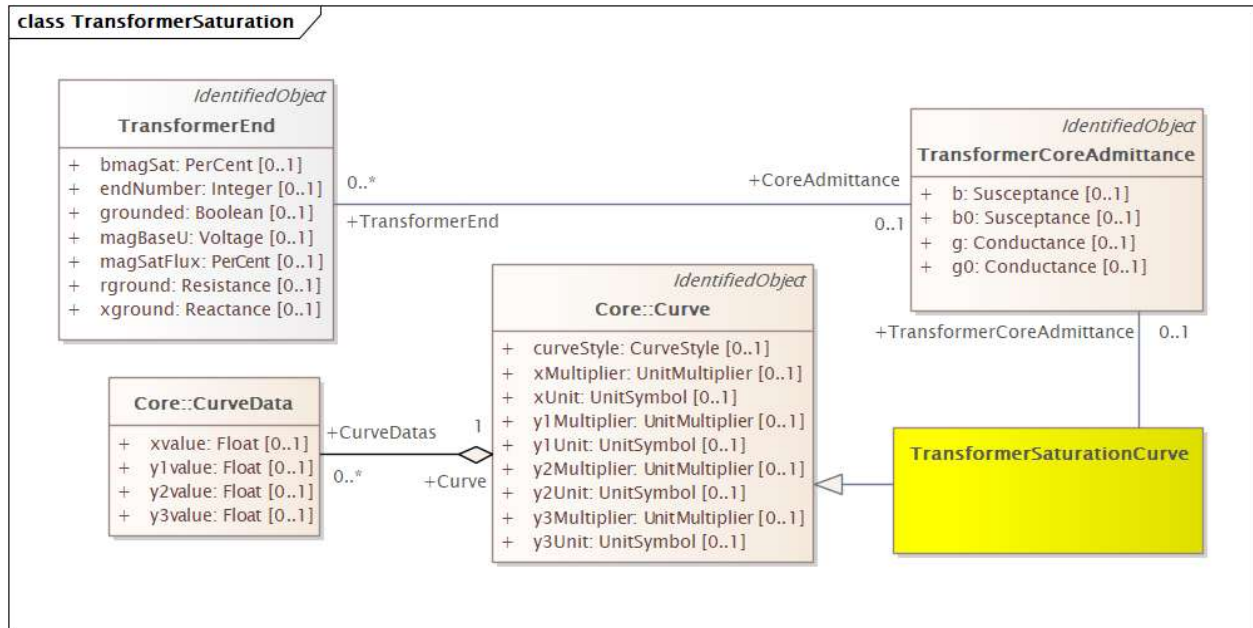


Figure 8: CIM extension UML for transformer saturation added to Wires:TransformerCoreAdmittance, new class in yellow.

CIM Extensions for Exact IBR Control Code

Figure 9 shows new IBR class associations on a UML diagram, with new attributes as defined in Table 2, to complete representation of the orange highlights in Figure 5 for CIM. Two of these, *IEEECigreDLL* and *InverterBasedResourcePlant*, are optional associations to a *PowerElectronicsConnection*, and they inherit from *PowerSystemResource* and *IdentifiedObject*. The *name* and *mRID* attributes of *IdentifiedObject* support unique and persistent identifiers for the new classes.

The *IEEECigreDLL* attributes in Table 2 can specify the location and vintage/version of a DLL, which is assumed to comply with a standard interface specification for the DLL. There should be no attempt to model details of the DLL interface in CIM. Instead, the specification should be incorporated by reference. Model builders and EMT tool vendors would be responsible for using the

specification to connect the DLL to the inverter bridge, and other components, at simulation time⁶. The class attributes include the model's expected *timeStep*, and its support for EMT and/or RMS (PSPDT) simulation modes. These *timeStep*, *supportsEMT*, and *supportsRMS* attributes are discoverable from the model's application program interface (API) at runtime, so they are not strictly necessary in the UML. However, they are useful attributes for the repository to determine whether a DLL is compatible with an EMT network simulation before launching it.

Table 2 includes a starting point for attributes of *InverterBasedResourcePlant*, primarily focused on the passive filter components in Figure 5. This list would be refined in collaboration with EMT tool vendors and other interested parties. For example, *switchingFrequency=0* implies use of an average source model, but the CIM working groups generally prefer a more explicit Boolean attribute for such choices. It's expected that a tool-neutral set of attributes can be developed for *InverterBasedResourcePlant*, based on experience with other use cases that were successfully represented in CIM.

Table 2: CIM Attributes for two new IBR Plant Classes

Name	CIM Type	Description
<i>IEEECigreDLL</i>		
vendorName	String	Name of the model provider
firmwareVersion	String	Version of the hardware provider's firmware
firmwareInstalled	DateTime	Date and time of the firmware installation at the IBR plant
uri	String	Location of the DLL, either a universal resource locator or network-accessible filename.
timeStep	Float	Hard-wired simulation time step for this DLL
supportsEMT	Boolean	True if the DLL supports EMT simulation
supportsRMS	Boolean	True if the DLL supports RMS simulation
<i>InverterBasedResourcePlant</i>		
dcLinkVoltage	Voltage	Voltage of the DC bus, if applicable
dcLinkCapacitance	Capacitance	Capacitance (plus-to-minus) of the DC bus, if applicable
acFilterRgrid	Resistance	Grid-side resistance of the AC tee filter
acFilterLgrid	Inductance	Grid-side inductance of the AC tee filter
acFilterCapacitance	Capacitance	Shunt capacitance of the AC tee filter
acFilterRbridge	Resistance	Inverter bridge-side resistance of the AC tee filter
acFilterLbridge	Inductance	Inverter bridge-side inductance of the AC tee filter
switchingFrequency	Frequency	PWM switching frequency

Four of the new classes in Figure 9, *IEEECigreDLLValue*, *IEEECigreDLLInteger*, *IEEECigreDLLFloat*, and *IEEECigreDLLString*, support the DLL parameter sets with saved values. The values may be one of the ten enumerated *IEEECigreDLLParameterKind* selections, and the *sequenceNumber* of each value must match the expected order in the model's API. The *sequenceNumber* and *parameterType* can be determined from the API at runtime, but including these in the UML allows the parameter values to be saved in the EMT model repository. The API also provides access to parameter names,

⁶ CIM for Dynamics already represents *RemoteSignals* as needed for *SynchronousMachine* control. Extensions may (or may not) be required for IBR plants, but the details are to be determined.

default values, minimum values, and maximum values, but these attributes are not essential to maintain the repository, so they are left out of the proposed UML. Each stored parameter value would be instantiated as one of the three concrete classes, indicated in Table 3. The classes for integer and float values can support different bit lengths, as directed by *parameterType*.

Table 3: Instantiation of IEEE Cigre DLL values in UML classes.

IEEECigreDLLParameterKind	Concrete Class Instantiation
Char_Val	IEEECigreDLLInteger
Char_Ptr	IEEECigreDLLString
Int8_Val	IEEECigreDLLInteger
Int16_Val	IEEECigreDLLInteger
Int32_Val	IEEECigreDLLInteger
UInt8_Val	IEEECigreDLLInteger
UInt16_Val	IEEECigreDLLInteger
UInt32_Val	IEEECigreDLLInteger
Real32_Val	IEEECigreDLLFloat
Real64_Val	IEEECigreDLLFloat

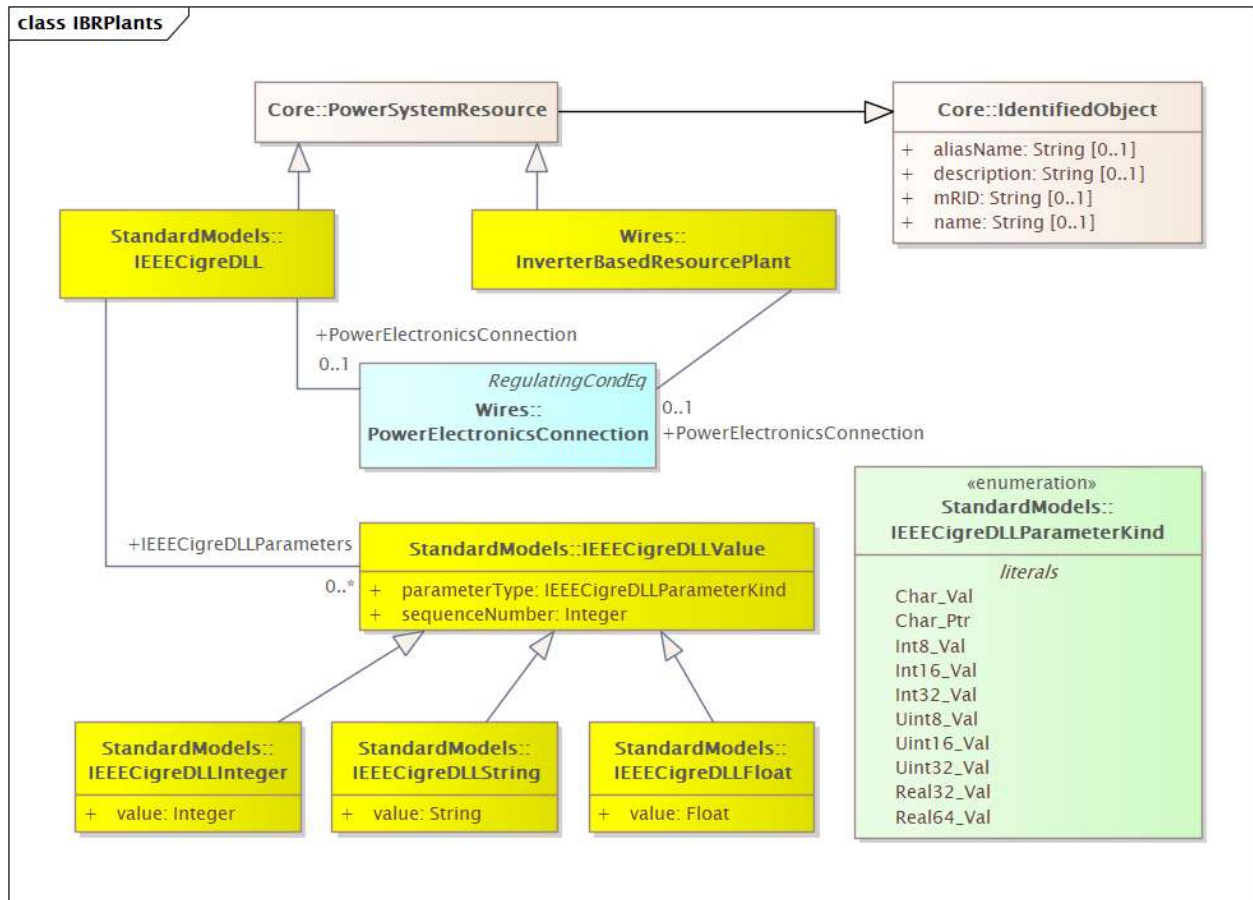


Figure 9: CIM extension UML for plant parameters and DLLs added to Wires:PowerElectronicsConnection. New classes are shown in yellow, and a new enumeration is shown in green.

The DLL API includes *ExternalInputs* and *ExternalOutputs* that must be connected to other parts of the network model. Some of that information could be represented in Figure 9, but that would not suffice for connecting these interface variables to other parts of the network model. Some of these inputs and outputs may be voltages and currents at network terminals, but others are not. It is up to the simulator, probably with user intervention, to make the right connections. For some use cases, the inputs and outputs could be further standardized, e.g., as in Figure 6. Future extensions of the UML may be developed to support this task.

Future Work for CIM Implementation

A CIM-based framework with automation can provide more benefits with future work:

- A CIM-based framework can support automated IBR model validation using scripts and SMIB test systems [18]. The support for different test systems and EMT tools should be broadened.
- Interoperability tests for PSPDT, SCKT, and EMT models should be sponsored by the CIM Users Group. These interop tests have been essential in fixing gaps and errors in the CIM UML, and in promoting or enabling acceptance by utilities and tool vendors.
- Adopt the standardization of asset-based information for transformers and transmission lines. Distribution utilities have already done much of this work through asset management and geographic information systems, which reduces the time to build and validate network models for distribution planning and DER interconnection studies.
- Adoption of node-breaker modeling for EMT and protection system design. This enables more accurate simulation of fault events and protection system response. The bus-branch model, currently used for many PF, PSPDT, and SCKT models, only represents the system state before and after contingencies.
- CIM fragments for IBR balance-of-plant models, shown in Figure 5, could help regularize the treatment of network subsystems. IEC Technical Committee 57 Working Group 13 has worked sporadically on this concept but has not completed it. IBR plant models provide another use case motivating the completion of this work on CIM fragments.
- Formalizing the DLL interface as an IEEE Standard will foster its adoption in CIM by reference, along with other benefits of standardization.

Future Work for IEEE Open-Source Implementation

It is technically feasible to implement an EMT model repository using the CIM as described above, but significant economic and organizational barriers exist:

- The CIM has not been widely adopted in North America, so it doesn't have the same critical mass as in Europe, where a legislative mandate drives model exchange for planning and operations. A relatively small number of North American transmission and distribution utilities have adopted CIM for other use cases, e.g., asset management, meter data management, etc. For transmission system analysis, a tool-specific text file format (TFF), which predates the CIM by several decades, has already achieved critical mass.

- Intellectual property conflicts have developed between the IEC (standards-making body) and the CIM Users Group (custodians of the data schema in UML, which EPRI originally contributed to the IEC). This increases the cost of development, with duplicate efforts by the organizations. It also creates vendor uncertainty about whether it is safe to use standard code components to implement the CIM.
- The cost of relevant IEC standards is higher than IEEE standards. For example, IEEE 2800 costs \$167 for the PDF. In the DER space, a complete bundle of IEEE 1547 costs \$500, and the two essential (“shall do”) parts cost \$367. The IEC standards for CIM Dynamics comprise about six parts, at a cost of 2465 CHF or about \$2700. In practice, it’s also necessary to join the CIM User’s Group for access to the most current UML. This price differential can be a barrier for smaller organizations.

The IEEE provides a relatively new mechanism for open-source projects, in which software may be developed and made available as part of the standards development process. Near the end of 2024, the Working Group for IEEE P1729, Recommended Practice for Electric Power Distribution System Analysis, added an open-source component for maintaining the IEEE Distribution Test Feeders, which had been maintained since 1991 on a variety of web sites and publications. The new open-source repository will formalize the maintenance of these electronic artifacts, including documentation and version control [19]. P1729 has chosen the Berkeley Software Distribution (BSD) 3-clause license, but the Apache 2.0 license could also be chosen. Experience to date has been good.

Therefore, the IEEE open-source platform provides an alternative to the CIM for the rapid development and adoption of practical EMT model repositories. The time frame could range from two to four years, especially if leveraging data formats that are already widely used in North America. Two separate projects are proposed. Both would be Tier 4 IEEE Open-Source projects, i.e., developed as part of companion IEEE Standards rather than standalone software.

1. IEEE Guide to Electromagnetic Transient Model Interoperability for Transmission Systems. A guide has “may do” strength in the ecosystem of IEEE Standards. The scope would include methods of using and extending the new EMT model repository but exclude PSPDT models and distribution systems. It would refer to the next project (as normative) on real-code modeling. Because CIM remains a viable alternative, this project is not proposed as “should do.” The TFF vendor would contribute to and hopefully participate in this project.⁷
2. IEEE Recommended Practice for Code Modeling in Electric Power Transmission and Distribution Systems. A recommended practice has “should do” strength in the ecosystem of IEEE Standards. Building on [16], this project could formalize inputs and outputs as in Figure 6. It could generalize the DLL specification for more use cases, e.g., time-series power flow simulations on distribution systems, non-IBR devices, run-time configuration,

⁷ Working group members may expect to spend up to eight hours each month (average) on virtual meetings and off-line contributions for up to four years. The TFF vendor may have one (or more) employees join the working group. It’s recommended, but not required, that participants be individual members of the IEEE Standards Association, so they can vote on the standard during balloting. This project will follow the individual process for IEEE standards, which means that working group members agree to exercise their own professional judgement. Members declare their affiliations, either an employer or a funding sponsor. In this project, the TFF vendor’s direct interests are to be met via the Contributor License Agreements [20].

etc. Alternatives exist for real-code and other code modeling⁸, but they are more suited for research or development work than for electric utility applications, so this project is proposed as “should do.”

A rough outline of the process for an EMT model interoperability guide follows:

1. Obtain buy-in from the vendor that defined the TFF already used in North America. EMT consultants and tool vendors already use TFF converters that they developed and maintain themselves. For this standardization proposal, buy-in means at minimum offering permission to document TFF in open-source electronic format, which a new working group would then maintain and extend separately from the original software product(s). This buy-in will save development time and increase acceptance of the new EMT network model.
2. Recruit about twenty working group members from various software tool vendors, consultants, utilities, and other stakeholders. The NERC EMT TF and the Oak Ridge National Laboratory’s annual EMT workshop offer some opportunities for recruitment. Contributors and participants need to execute Contributor License Agreements (CLA) with IEEE [20]. The Apache 2.0 License and CLA help contributors preserve their intellectual property rights.
3. Write an IEEE Project Authorization Request (PAR), including the open-source component, to develop a Guide under sponsorship of the IEEE PES Analytical Methods in Power Systems (AMPS) Committee. AMPS is responsible for EMT within the IEEE PES Technical Committee structure. This kicks off the IEEE process with in-person and virtual meetings, document and software development, balloting and comment resolution, etc.⁹
4. The new schema could be documented from an existing JavaScript Object Notation (JSON) version of TFF [22]. Compared to the extensible markup language (XML) files used in CIM, the JSON files are easier to work with. Many programming languages, scripting languages, and open-source tools support JSON.
5. Technical work begins with automated generation of JSON Schema from the TFF. This roughly corresponds to the data modeling already in CIM with Dynamics. The extensions for transformer saturation, IBR plants, and DLL interfaces would be added as discussed above for CIM.¹⁰ Consultants and tool vendors may contribute file format converters and test cases.
6. This effort would publish a JSON Schema file, which defines the name, type, format, range, default value, description, and relationships for all attributes in the schema. Documentation, diagrams, and code files are built from the JSON Schema file using

⁸ MATLAB/Simulink plays an important role in vendor control code verification. Languages like VHDL-AMS (IEEE 1076), Verilog-A and MAST have been used in other domains, e.g., automotive, mixed-signal circuits. The functional mockup interface (FMI), a Modelica Association project, has over two hundred supporting tools [21]. At least two of those FMI-supporting tools are used by electric utilities for planning and design work, i.e., EMTP[®] and PowerFactory. Although the DLL interface has more vendor support within the utility industry, FMI could become a viable alternative with more features. The Recommended Practice should compare FMI to the DLL interface and identify situations where FMI could present a viable alternative.

⁹ In parallel, write a PAR for a Recommended Practice (RP) on real-code modeling, also sponsored by AMPS. The EMT model repository Guide will refer to RP. The working group memberships will probably overlap.

¹⁰ It may be desirable to incorporate other selected CIM ideas, e.g., bus-branch and node-breaker topology, bus naming, unique identifiers, SI units, etc. Line constants input may also be developed as future extensions. However, these extensions should be deferred to encourage rapid development and adoption of the core schema.

appropriate tools, of which there are many options. JSON example files may be validated against the JSON Schema.

7. The IEEE working group, collaborating with or including the TFF vendor, should maintain compatibility between the EMT repository schema and the TFF software product. This allows both schemas to evolve along different paths at different paces. The life cycle of an IEEE Standards project is ten years, so the TFF product will be updated more frequently. The standard and JSON Schema should be written to allow for extensions and versions. This would simplify the process of maintaining the standard on ten-year cycles. If the standard and TFF software product fall out of synchronism, two options should close the gaps. First, customer-driven support for legacy file export may be included in the TFF software vendor's product. Second, if that does not occur, then interested parties may write their own version translators using the published JSON Schema.
8. Other software vendors, consultants, utilities, and stakeholders would implement their own import and export converters using the EMT repository schema. The published JSON Schema, documentation and open-source license would allow this activity to occur safely.

Currently, some of this TFF interface work already occurs ad hoc by various industry stakeholders. Without standards, this ad hoc work is duplicative and prone to version mismatch or other errors. Ad hoc extensions for the DLL interface, IBR plant parameters, transformer saturation, etc., will not be portable, especially if they are not required in the PF and PSPDT use cases. Standardization would address all these barriers for the EMT stakeholders. Standardization can also lead to demand for new products, benefiting the TFF vendor and others.

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Scope from EMT TF WI3 Meeting Minutes

White paper “Resources for EMT modeling of bulk power systems” (Tom M. responsible person: due by end of Q3 2024)

Abstract: *With EMT studies, there is not always a one size fits all representation for modeling general power system equipment. Many of the commercially available tools which are used for automated creation of EMT models have a default method of modeling equipment and will generate usable models. For example, these tools will typically import steady-state and dynamics data from a phasor domain tool and will generate an EMT model that can run time domain simulations at a given simulation time-step. However, because of limitations of data available in the source databases, such models will not include many system modeling details that are typically important for EMT level simulation, such as:*

- *Correct zero sequence impedance of transmission lines or cables*
- *Frequency dependent impedance of transmission lines or cables*
- *Mutual coupling between transmission lines*
- *Transformer winding configuration and grounding information*
- *Transformer saturation characteristics*
- *Custom or user-defined representation for load or generation*

It is necessary for the study engineer to ensure that power system equipment is modeled appropriately for the phenomena of interest under evaluation. Providing a complete and detailed discussion on power system modeling for EMT is outside the scope of this document.

Topics to cover include:

- a. Real code models for IBRs
- b. Resources for obtaining “missing” EMT model data e.g.,
 - i. Transformer winding configuration and grounding information
 - ii. Transformer saturation characteristics
 - What are rules of thumb/guidance for determining saturation characteristics? (e.g., selection of knee point, selection of excitation current, selection of air-core reactance)
- c. When and why is the missing data important? For example, do you always need frequency dependent transmission lines?

Revision History

#	Date	Comments
1	9/27/2024	Initial draft with background and CIM extensions
2	1/31/2025	Add parameter sets to DLL interface, add IEEE open-source project proposal
3	3/13/2025	Cigre TB published, JSON options, DLL inputs/outputs, Apache license/CLA. Separate IEEE open-source project for DLL models.
4	3/20/2025	FMI alternative, added HVDC-wise reference.

See [23] for the latest version.